1. **Resolve Merge Conflicts**
    - Create a merge conflict intentionally (two users editing the same line).
    - Resolve the conflict and push the changes.

```
nothing to commit, working tree clean
sharafkamaal@192 kamaal123 % git branch --list
  hotfix
* main
sharafkamaal@192 kamaal123 % vi file.txt
sharafkamaal@192 kamaal123 % cat file.txt
echo "Welcome to Git - User 2"
sharafkamaal@192 kamaal123 %
```

```
sharafkamaal@192 kamaal123 % git checkout release
Switched to branch 'release'
sharafkamaal@192 kamaal123 % git branch --list
  hotfix
  main
* release
sharafkamaal@192 kamaal123 % vi file.txt
sharafkamaal@192 kamaal123 % cat file.txt
echo "Welcome to Git - User 1"
sharafkamaal@192 kamaal123 %
```

```
~/devops-techiehorizon/kamaal123/kamaal123 — root@ip-172-31-31-201:/kamaal/scripts — -zsh
~/devops-techiehorizon — -zsh
sharafkamaal@192 kamaal123 % git checkout main
Switched to branch 'main'
Your branch and 'origin/main' have diverged,
and have 1 and 4 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch and 'origin/main' have diverged,
and have 1 and 4 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file.txt

sharafkamaal@192 kamaal123 % git commit -m "checking merge conflict1"
[main 25b63cc] checking merge conflict1
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
sharafkamaal@192 kamaal123 % git checkout release
Switched to branch 'release'
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git status
On branch release
nothing to commit, working tree clean
sharafkamaal@192 kamaal123 % vi file.txt
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git status .
On branch release
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file.txt

sharafkamaal@192 kamaal123 % git commit -m "checking merge conflict on release branch"
[release 5feec34] checking merge conflict on release branch
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
sharafkamaal@192 kamaal123 %
```

```
<<<<<<< HEAD
echo "Welcome to Git - User1"
=======
echo "Welcome to Git - User 1"
>>>>>>> main
~
~
~
~
~
~
~
~
```

```
echo "Welcome to Git - User1 user2"
~
~
~
~
~
~
~
~
```

```
[sharafkamaal@192 kamaal123 % vi file.txt
[sharafkamaal@192 kamaal123 % cat file.txt
 echo "Welcome to Git - User1 user2"
[sharafkamaal@192 kamaal123 % git merge main
 error: Merging is not possible because you have unmerged files.
 hint: Fix them up in the work tree, and then use 'git add/rm <file>'
 hint: as appropriate to mark resolution and make a commit.
 fatal: Exiting because of an unresolved conflict.
[sharafkamaal@192 kamaal123 % git add .
[sharafkamaal@192 kamaal123 % git commit -m "resolve the conflict issue"
 [release bfc9ca6] resolve the conflict issue
[sharafkamaal@192 kamaal123 % git merge main
 Already up to date.
 sharafkamaal@192 kamaal123 %
```

2.Recover Deleted Branch

- Delete a local branch and then recover it using the reflog.
-

```
[sharafkamaal@192 kamaal123 % git switch hotfix
 Switched to branch 'hotfix'
 Your branch is up to date with 'origin/hotfix'.
[sharafkamaal@192 kamaal123 % ls
 25              54              file1.txt       file2.txt       password        README.md
[sharafkamaal@192 kamaal123 % vi file2.txt
[sharafkamaal@192 kamaal123 % cat file2.txt
 echo "Feature branch content"
 sharafkamaal@192 kamaal123 %
```

```
sharafkamaal@192 kamaal123 % git branch -d hotfix
warning: deleting branch 'hotfix' that has been merged to
         'refs/remotes/origin/hotfix', but not yet merged to HEAD
Deleted branch hotfix (was 241bc05).
sharafkamaal@192 kamaal123 % git branch --list
* main
  release
sharafkamaal@192 kamaal123 %
```

```
sharafkamaal@192 kamaal123 % git reflog
25b63cc (HEAD -> main) HEAD@{0}: checkout: moving from hotfix to main
241bc05 (origin/hotfix) HEAD@{1}: checkout: moving from release to hotfix
bfc9ca6 (release) HEAD@{2}: commit (merge): resolve the conflict issue
5feec34 HEAD@{3}: commit: checking merge conflict on release branch
9d940b4 HEAD@{4}: checkout: moving from main to release
25b63cc (HEAD -> main) HEAD@{5}: commit: checking merge conflict1
9d940b4 HEAD@{6}: checkout: moving from release to main
9d940b4 HEAD@{7}: checkout: moving from main to release
9d940b4 HEAD@{8}: checkout: moving from release to main
9d940b4 HEAD@{9}: checkout: moving from main to release
9d940b4 HEAD@{10}: rebase (finish): returning to refs/heads/main
9d940b4 HEAD@{11}: rebase (squash): Creating two files
4035309 HEAD@{12}: rebase (squash): # This is a combination of 3 commits.
b972eeb HEAD@{13}: rebase (squash): # This is a combination of 2 commits.
d716abc HEAD@{14}: rebase (start): checkout HEAD~4
3113149 HEAD@{15}: commit (amend): Revert "doing some changes"
5f0e43d HEAD@{16}: reset: moving to HEAD
5f0e43d HEAD@{17}: revert: Revert "adding a file"
8f1f5a0 (origin/main, origin/HEAD) HEAD@{18}: reset: moving to HEAD~1
3a4f56e HEAD@{19}: reset: moving to 3a4f56
3a4f56e HEAD@{20}: reset: moving to 3a4f56
3a4f56e HEAD@{21}: reset: moving to 3a4f56
3a4f56e HEAD@{22}: reset: moving to HEAD
3a4f56e HEAD@{23}: commit: reseting the commit
8f1f5a0 (origin/main, origin/HEAD) HEAD@{24}: pull --rebase origin main (finish): returning to refs/heads/main
8f1f5a0 (origin/main, origin/HEAD) HEAD@{25}: pull --rebase origin main (pick): hiding password file
04b63f8 HEAD@{26}: pull --rebase origin main (start): checkout 04b63f805631e2c5614f426ef92dada75af52b94
c302fda HEAD@{27}: commit: hiding password file
241bc05 (origin/hotfix) HEAD@{28}: merge hotfix: Fast-forward
d716abc HEAD@{29}: checkout: moving from main to main
d716abc HEAD@{30}: checkout: moving from hotfix to main
241bc05 (origin/hotfix) HEAD@{31}: commit: adding a file
d716abc HEAD@{32}: clone: from https://github.com/sharafkamaal/kamaal123.git
sharafkamaal@192 kamaal123 % git branch hotfix 241bc05
sharafkamaal@192 kamaal123 % git branch --list
  hotfix
* main
  release
```

```
sharafkamaal@192 kamaal123 % git switch hotfix
M       file2.txt
Switched to branch 'hotfix'
sharafkamaal@192 kamaal123 % ls
25              54              file1.txt       file2.txt       password        README.md
sharafkamaal@192 kamaal123 %
```

3.Undo Wrong Push

- Push a wrong commit to GitHub, then undo it without losing history.

```
sharafkamaal@192 kamaal123 % git switch main
M       file2.txt
Switched to branch 'main'
Your branch and 'origin/main' have diverged,
and have 2 and 4 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)
sharafkamaal@192 kamaal123 % vi app.txt
sharafkamaal@192 kamaal123 % cat app.txt
echo "Wrong information"
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git staus
git: 'staus' is not a git command. See 'git --help'.

The most similar command is
        status
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch and 'origin/main' have diverged,
and have 2 and 4 different commits each, respectively.
  (use "git pull" if you want to integrate the remote branch with yours)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   app.txt
        modified:   file2.txt
```

```
sharafkamaal@192 kamaal123 % git rebase origin main
Successfully rebased and updated refs/heads/main.
sharafkamaal@192 kamaal123 % git push -u origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 10 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (10/10), 938 bytes | 938.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/sharafkamaal/kamaal123.git
   8f1f5a0..6146424  main -> main
branch 'main' set up to track 'origin/main'.
sharafkamaal@192 kamaal123 %
```

```
commit 6146424f8ec5e1c562767f320f7804e1cc37ad9f (HEAD -> main, origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:29:39 2026 +0530

    wrong commit
```

```
sharafkamaal@192 kamaal123 % git log
commit a9e286a8267aa56d9bdce880013a2eeb9715f987 (HEAD -> main, origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:31:57 2026 +0530

    Revert "wrong commit"
    echo "correcting the fix"
    This reverts commit 6146424f8ec5e1c562767f320f7804e1cc37ad9f.
```

4.Amend a Commit

- Make a commit, then add a missing file to it using git commit --amend.

step 1: we create 2 files
1. kite file
2 remote file

```
sharafkamaal@192 kamaal123 % vi kite
sharafkamaal@192 kamaal123 % cat kite
echo "Hello World"
sharafkamaal@192 kamaal123 % vi remote
sharafkamaal@192 kamaal123 % cat kite
echo "Hello World"
sharafkamaal@192 kamaal123 % cat remote
echo "This is the REMOTE"
sharafkamaal@192 kamaal123 %
```

-
- step 2 : I only add kite file using git add kite.

```
[sharafkamaal@192 kamaal123 % git add kite
[sharafkamaal@192 kamaal123 % git status
 On branch main
 Your branch is up to date with 'origin/main'.

 Changes to be committed:
   (use "git restore --staged <file>..." to unstage)
         new file:   kite
```

step 3: we commit the file using git commit -m “”

```
[sharafkamaal@192 kamaal123 % git commit -m "doing ammend task"
[main f85a223] doing ammend task
 1 file changed, 1 insertion(+)
 create mode 100644 kite
```

step 4:but we accidentally forgot to include remote file then we add it using git add remote

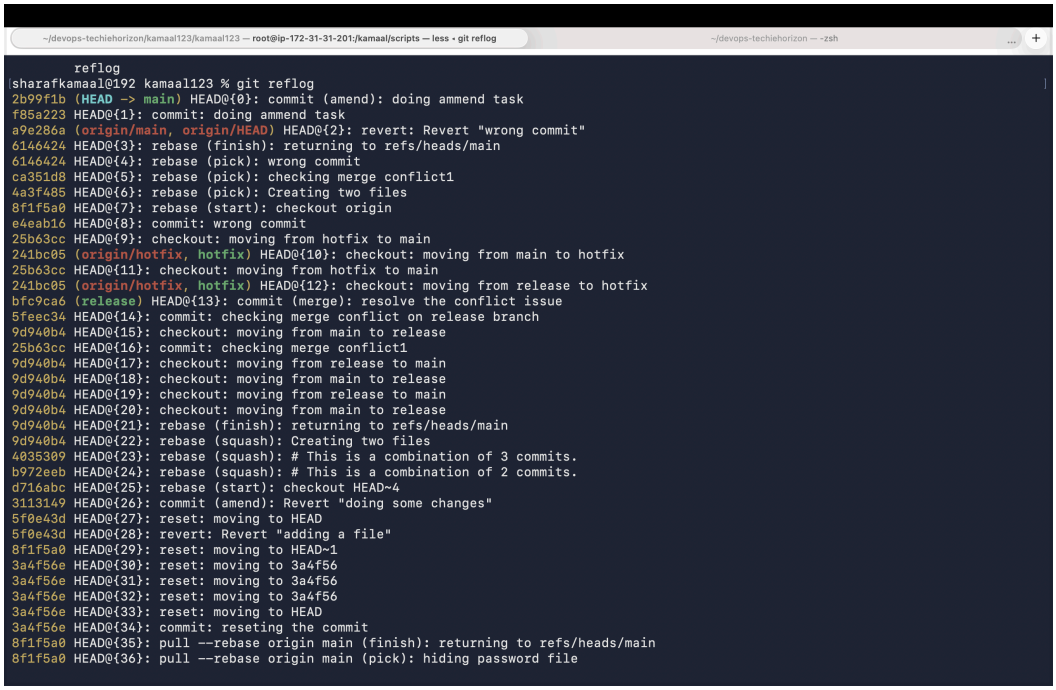
```
[sharafkamaal@192 kamaal123 % git add remote
[sharafkamaal@192 kamaal123 % git status
 On branch main
 Your branch is ahead of 'origin/main' by 1 commit.
   (use "git push" to publish your local commits)

 Changes to be committed:
   (use "git restore --staged <file>..." to unstage)
         new file:   remote
```

step 5: we commit the same file with same commit id using git commit —amend —no-edit

```
[sharafkamaal@192 kamaal123 % git commit --amend --no-edit
 [main 2b99f1b] doing ammend task
  Date: Mon Oct 26 12:46:43 2026 +0530
  2 files changed, 2 insertions(+)
  create mode 100644 kite
  create mode 100644 remote
 sharafkamaal@192 kamaal123 %
```

```
sharafkamaal@192 kamaal123 % git show --stat HEAD
commit 2b99f1bbbd9249bd6fb7e9ada3766882dd9a1d79 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:46:43 2026 +0530

    doing ammend task

 kite   | 1 +
 remote | 1 +
 2 files changed, 2 insertions(+)
sharafkamaal@192 kamaal123 %
```

5.Cherry-pick a Commit

- Take a specific commit from one branch and apply it to another branch.
  step1: we use git reflog so its shows the history of commit and we copy commit id
  - 

step2:we switch to main using git switch main

```
sharafkamaal@192 kamaal123 % git switch main
Already on 'main'
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```

step3: we copy the commit id and use the command git cherry-pick and commit id “”

```
sharafkamaal@192 kamaal123 % git switch release
fatal: cannot switch branch while cherry-picking
Consider "git cherry-pick --quit" or "git worktree add".
sharafkamaal@192 kamaal123 % git switch release
fatal: cannot switch branch while cherry-picking
Consider "git cherry-pick --quit" or "git worktree add".
sharafkamaal@192 kamaal123 % git cherry-pick --quit
sharafkamaal@192 kamaal123 % git switch release
file.txt: needs merge
error: you need to resolve your current index first
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Unmerged paths:
  (use "git restore --staged <file>..." to unstage)
  (use "git add <file>..." to mark resolution)
        both added:      file.txt

no changes added to commit (use "git add" and/or "git commit -a")
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git commit -m "cherypick"
[main cb879dc] cherypick
 1 file changed, 4 insertions(+)
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
sharafkamaal@192 kamaal123 % ls
25              alpha           file1.txt       kite            README.md
54              file.txt        file2.txt       password        remote
sharafkamaal@192 kamaal123 %
```

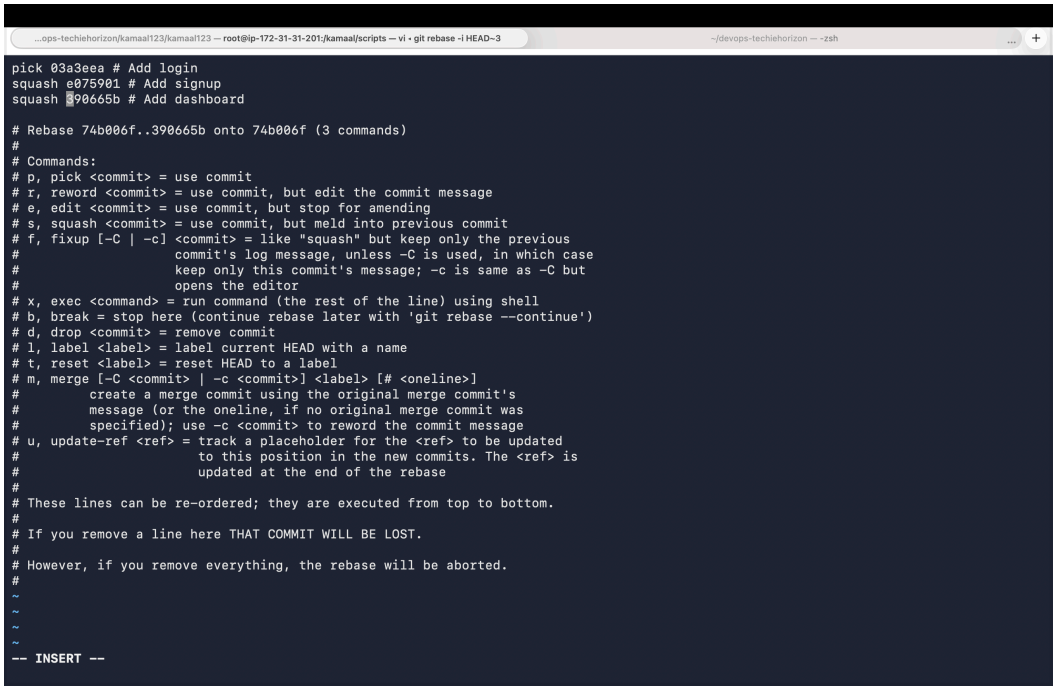
```
sharafkamaal@192 kamaal123 % git cherry-pick 5f0e43d
[main 74b006f] Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
 Date: Fri Oct 23 16:31:00 2026 +0530
 2 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 25
 delete mode 100644 54
sharafkamaal@192 kamaal123 % ls
alpha           file.txt        file1.txt       file2.txt       kite            password        README.md       remote
sharafkamaal@192 kamaal123 %
```

6.Interactive Rebase

- Reorder and squash multiple commits into a single clean commit.

step1 : first we create 3 files 1.login 2.dashboard 3.signup
step2:git add . git commit -m for every file
step3: we use git rebase -i HEAD~3 for multiple commits into a single clean commit.
step 4: Its open the editor and show pick then we have to replace pick with squash command and save it.

```
sharafkamaal@192 kamaal123 % git add login.txt
sharafkamaal@192 kamaal123 % git commit -m "Add login"
[main 03a3eea] Add login
 1 file changed, 1 insertion(+)
 create mode 100644 login.txt
sharafkamaal@192 kamaal123 % git add signup.txt
git commit -m "Add signup"
[main e075901] Add signup
 1 file changed, 1 insertion(+)
 create mode 100644 signup.txt
sharafkamaal@192 kamaal123 % git add dashboard.txt
git commit -m "Add dashboard"
[main 390665b] Add dashboard
 1 file changed, 1 insertion(+)
 create mode 100644 dashboard.txt
sharafkamaal@192 kamaal123 % git log --oneline -4
390665b (HEAD -> main) Add dashboard
e075901 Add signup
03a3eea Add login
74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
sharafkamaal@192 kamaal123 % git rebase -i HEAD~3
[detached HEAD d05c297] Add login
 Date: Mon Oct 26 14:51:53 2026 +0530
 3 files changed, 3 insertions(+)
 create mode 100644 dashboard.txt
 create mode 100644 login.txt
 create mode 100644 signup.txt
Successfully rebased and updated refs/heads/main.
sharafkamaal@192 kamaal123 %
```




```
sharafkamaal@192 kamaal123 % git log --oneline -4
d05c297 (HEAD -> main) Add login
74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
cb879dc cherypick
2b99f1b doing ammend task
sharafkamaal@192 kamaal123 % ls
alpha           file.txt        file2.txt       login.txt       README.md       signup.txt
dashboard.txt   file1.txt       kite            password        remote
sharafkamaal@192 kamaal123 % git rebase -i HEAD~3
Successfully rebased and updated refs/heads/main.
sharafkamaal@192 kamaal123 %
```

7.Tagging & Release

- Create a version tag (v1.0), push it to GitHub, then delete and restore it.

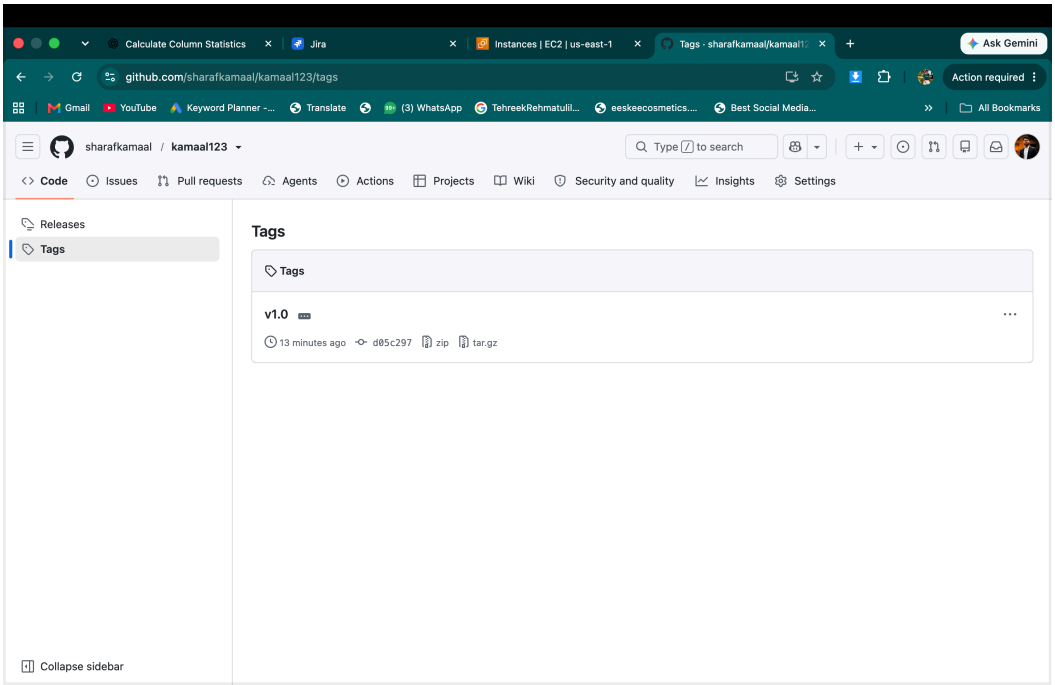
step1: we create version tag (v1.0) using git tag v1.0

- step2:for checking we use git tag, git show for commit hash

```
sharafkamaal@192 kamaal123 % git tag v1.0
sharafkamaal@192 kamaal123 % git tag
v1.0
sharafkamaal@192 kamaal123 % git show v1.0
commit d05c297152671788f38fbfd38a25154077f0e1db (HEAD -> main, tag: v1.0)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 14:51:53 2026 +0530

    Add login

    Add signup

    Add dashboard

diff --git a/dashboard.txt b/dashboard.txt
new file mode 100644
index 0000000..fa10263
--- /dev/null
+++ b/dashboard.txt
@@ -0,0 +1 @@
+Dashboard feature
diff --git a/login.txt b/login.txt
new file mode 100644
index 0000000..f00f155
--- /dev/null
+++ b/login.txt
@@ -0,0 +1 @@
+Login feature
diff --git a/signup.txt b/signup.txt
new file mode 100644
index 0000000..bab800c
--- /dev/null
+++ b/signup.txt
```

- step3: push the code in git hub and check in github account

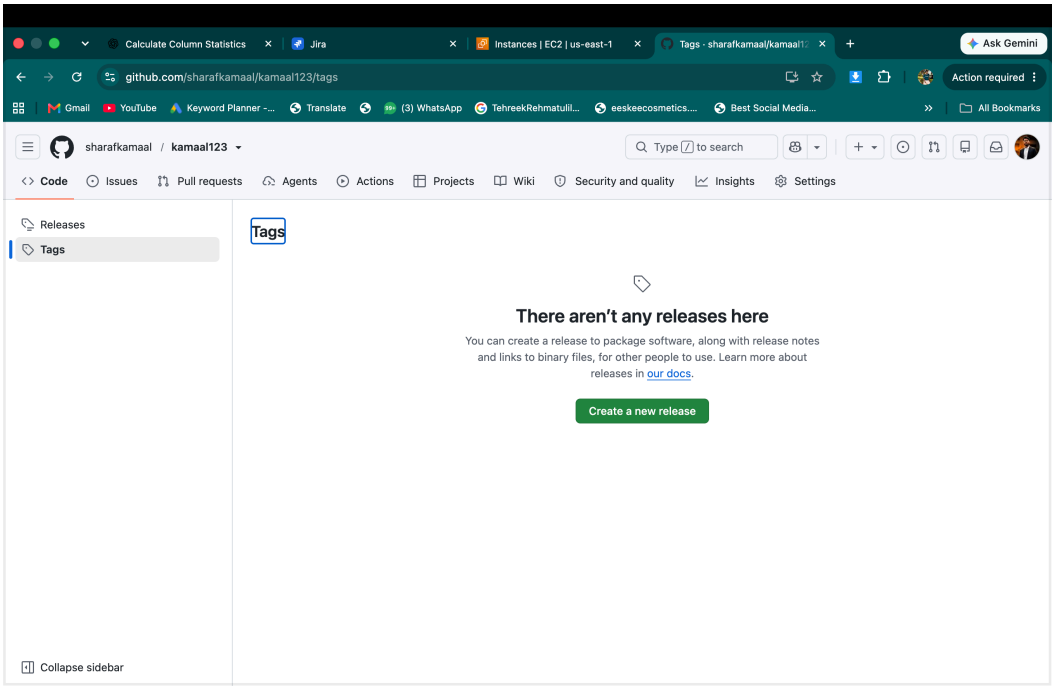
```
nothing to commit, working tree clean
[sharafkamaal@192 kamaal123 % git push origin v1.0
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 10 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (14/14), 1.36 KiB | 1.36 MiB/s, done.
Total 14 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/sharafkamaal/kamaal123.git
 * [new tag]         v1.0 -> v1.0
[sharafkamaal@192 kamaal123 % git tag -d v1.0
Deleted tag 'v1.0' (was d05c297)
[sharafkamaal@192 kamaal123 % git tag
[sharafkamaal@192 kamaal123 % git push origin --delete v1.0
To https://github.com/sharafkamaal/kamaal123.git
 - [deleted]         v1.0
sharafkamaal@192 kamaal123 %
```



step 4: we delete this tag using git tag -d v1.0
step5: then we push the code after deleting it

```
[sharafkamaal@192 kamaal123 % git tag -d v1.0
Deleted tag 'v1.0' (was d05c297)
[sharafkamaal@192 kamaal123 % git tag
[sharafkamaal@192 kamaal123 % git push origin --delete v1.0
To https://github.com/sharafkamaal/kamaal123.git
 - [deleted]         v1.0
[sharafkamaal@192 kamaal123 % git tag v1.0 d05c2
[sharafkamaal@192 kamaal123 % git tag
v1.0
[sharafkamaal@192 kamaal123 % git push origin v1.0
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 10 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (14/14), 1.36 KiB | 1.36 MiB/s, done.
Total 14 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/sharafkamaal/kamaal123.git
 * [new tag]         v1.0 -> v1.0
sharafkamaal@192 kamaal123 %
```

step6: then check it in github account in tags



8.Clone with Sparse Checkout

- Clone only a subdirectory of a repo using sparse checkout.

```
[sharafkamaal@192 devops-techiehorizon % git clone --no-checkout https://github.com/netology-code/py-homework-basic-files.git
Cloning into 'py-homework-basic-files'...
remote: Enumerating objects: 92, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 92 (delta 7), reused 3 (delta 3), pack-reused 78 (from 1)
Receiving objects: 100% (92/92), 95.93 KiB | 749.00 KiB/s, done.
Resolving deltas: 100% (22/22), done.
[sharafkamaal@192 devops-techiehorizon % git sparse-checkout init cone
[sharafkamaal@192 devops-techiehorizon % git sparse-checkout set 6.classes
```

```
sharafkamaal@192 devops-techiehorizon % cd py-homework-basic-files
sharafkamaal@192 py-homework-basic-files % ls
sharafkamaal@192 py-homework-basic-files % git sparse-checkout init cone
sharafkamaal@192 py-homework-basic-files % git sparse-checkout set 6.classes
sharafkamaal@192 py-homework-basic-files % ls
sharafkamaal@192 py-homework-basic-files % git switch main
fatal: invalid reference: main
sharafkamaal@192 py-homework-basic-files % git switch master
Already on 'master'
Your branch is up to date with 'origin/master'.
sharafkamaal@192 py-homework-basic-files % ls
6.classes       README.md
sharafkamaal@192 py-homework-basic-files %
```

9.Reset vs Revert Challenge

- Demonstrate the difference between git reset --hard and git revert in a repo.

```
sharafkamaal@192 html2canvas % cd ..
sharafkamaal@192 kamaal123 % git log
commit d05c297152671788f38fbfd38a25154077f0e1db (HEAD -> main, tag: v1.0)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 14:51:53 2026 +0530

    Add login

    Add signup

    Add dashboard

commit 74b006f33012eb68d709a8464eb1b7cc12083974
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:31:00 2026 +0530

    Revert "adding a file"
    "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

commit cb879dc24d291e08a6963382a4a73cce8290ee3f
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 14:45:56 2026 +0530

    cherypick

commit 2b99f1bbbd9249bd6fb7e9ada3766882dd9a1d79
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:46:43 2026 +0530

    doing ammend task

commit a9e286a8267aa56d9bdce880013a2eeb9715f987 (origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:31:57 2026 +0530
```

-

```
sharafkamaal@192 kamaal123 % git reset --hard HEAD~1
HEAD is now at 74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
sharafkamaal@192 kamaal123 % git log
commit 74b006f33012eb68d709a8464eb1b7cc12083974 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:31:00 2026 +0530

    Revert "adding a file"
    "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

commit cb879dc24d291e08a6963382a4a73cce8290ee3f
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 14:45:56 2026 +0530

    cherypick

commit 2b99f1bbbd9249bd6fb7e9ada3766882dd9a1d79
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:46:43 2026 +0530

    doing ammend task

commit a9e286a8267aa56d9bdce880013a2eeb9715f987 (origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 12:31:57 2026 +0530

    Revert "wrong commit"
    echo "correcting the fix"
```

```
sharafkamaal@192 kamaal123 % git log
commit 8ccbe41ec15a2f37274e8f81851865c4027ec755 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Mon Oct 26 15:56:09 2026 +0530

    Reapply "adding a file"

    what a joke
    his reverts commit 74b006f33012eb68d709a8464eb1b7cc12083974.

commit 74b006f33012eb68d709a8464eb1b7cc12083974
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:31:00 2026 +0530

    Revert "adding a file"
    "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
```

10.Detached HEAD Challenge

- Checkout a specific commit (detached HEAD state) and create a new branch from it.

```
sharafkamaal@192 kamaal123 % git checkout 8ccbe4
Note: switching to '8ccbe4'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 8ccbe41 Reapply "adding a file"
sharafkamaal@192 kamaal123 % git status
HEAD detached at 8ccbe41
Untracked files:
```

-

```
sharafkamaal@192 kamaal123 % git switch -c feature-from-8ccbe41
Switched to a new branch 'feature-from-8ccbe41'
sharafkamaal@192 kamaal123 % git switch -c feature-from-8ccbe41
fatal: a branch named 'feature-from-8ccbe41' already exists
sharafkamaal@192 kamaal123 % git switch -c feature-from-8ccbe41
fatal: a branch named 'feature-from-8ccbe41' already exists
sharafkamaal@192 kamaal123 % git branch
* feature-from-8ccbe41
  hotfix
  main
  release
sharafkamaal@192 kamaal123 %
```

-

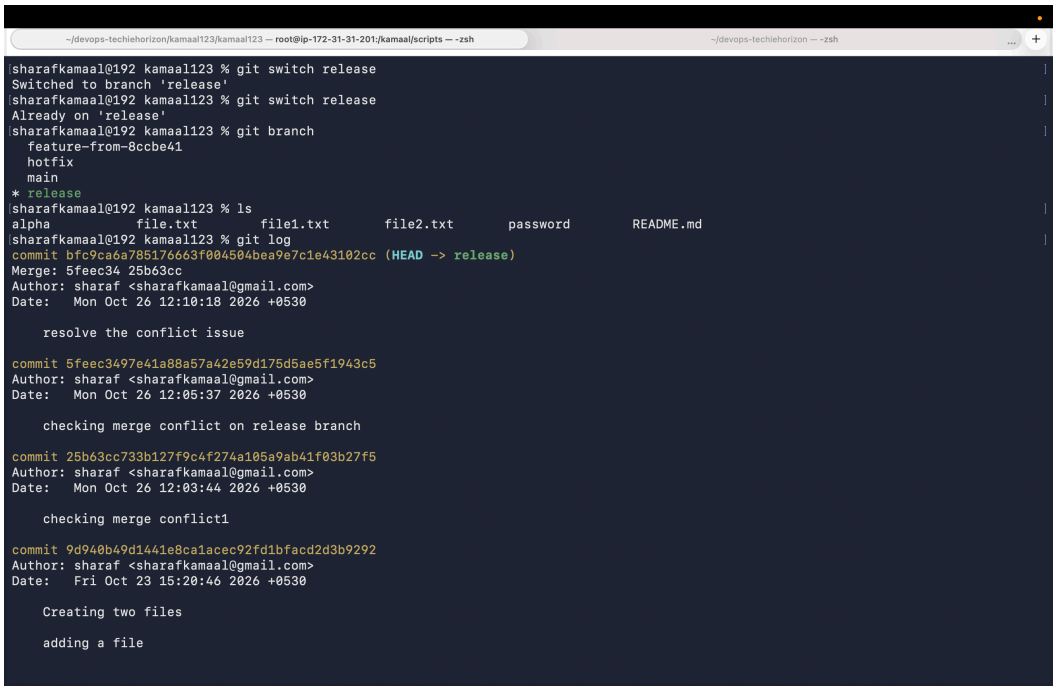
```
sharafkamaal@192 kamaal123 % git push -u origin feature-from-8ccbe41
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 268 bytes | 268.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature-from-8ccbe41' on GitHub by visiting:
remote:      https://github.com/sharafkamaal/kamaal123/pull/new/feature-from-8ccbe41
remote:
To https://github.com/sharafkamaal/kamaal123.git
 * [new branch]      feature-from-8ccbe41 -> feature-from-8ccbe41
branch 'feature-from-8ccbe41' set up to track 'origin/feature-from-8ccbe41'.
sharafkamaal@192 kamaal123 %
```

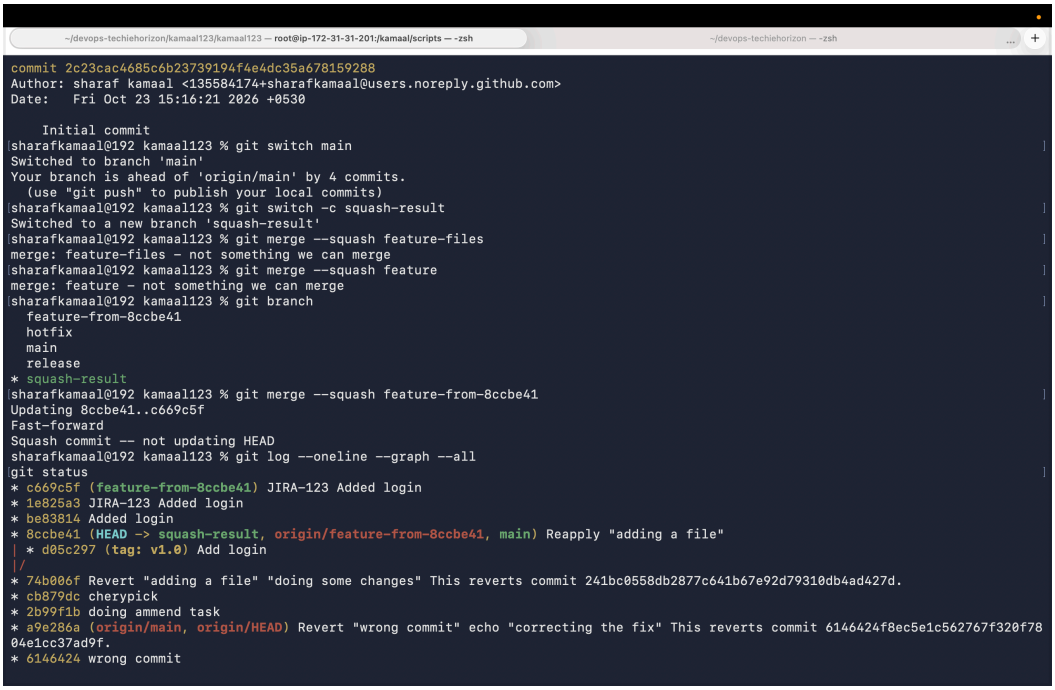
11.Git Hooks Challenge

- Configure a pre-commit hook to reject commits without a message format (e.g., must start with JIRA-XXX).

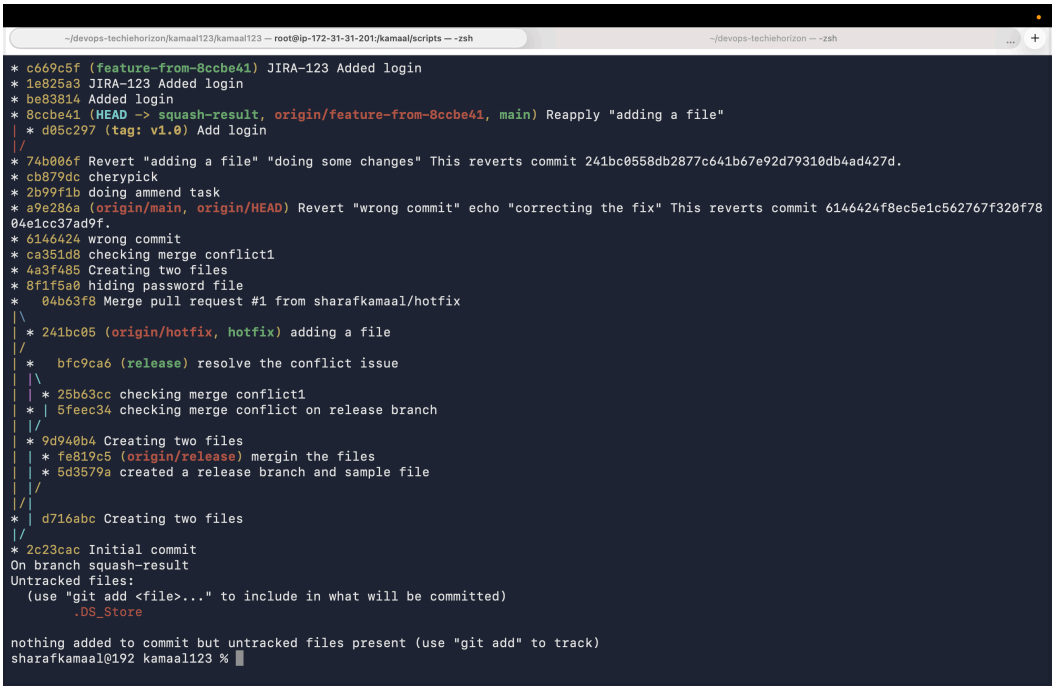
```
sharafkamaal@192 kamaal123 % vi .git/hooks/commit-msg
sharafkamaal@192 kamaal123 % cat .git/hooks/commit-msg
#!/bin/sh

message=$(cat "$1")

if ! echo "$message" | grep -Eq '^JIRA-[0-9]+ '; then
    echo "Commit rejected!"
    echo "Use: JIRA-123 Your message"
    exit 1
fi

echo "Commit message accepted!"
exit 0
sharafkamaal@192 kamaal123 % chmod 755 .git/hooks/commit-msg
sharafkamaal@192 kamaal123 % git commit --allow-empty -m "Added login"
Commit rejected!
Use: JIRA-123 Your message
sharafkamaal@192 kamaal123 % git commit --allow-empty -m "JIRA-123 Added login"
Commit message accepted!
[feature-from-8ccbe41 c669c5f] JIRA-123 Added login
sharafkamaal@192 kamaal123 %
```

12.Squash Merge vs Rebase Merge

- Show the difference between squash merge and rebase merge with evidence.
- 




```
commit 2c23cac4685c6b23739194f4e4dc35a678159288
Author: sharaf kamaal <135584174+sharafkamaal@users.noreply.github.com>
Date:   Fri Oct 23 15:16:21 2026 +0530

    Initial commit
sharafkamaal@192 kamaal123 % git switch main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)
sharafkamaal@192 kamaal123 % git switch -c squash-result
Switched to a new branch 'squash-result'
sharafkamaal@192 kamaal123 % git merge --squash feature-files
merge: feature-files - not something we can merge
sharafkamaal@192 kamaal123 % git merge --squash feature
merge: feature - not something we can merge
sharafkamaal@192 kamaal123 % git branch
  feature-from-8ccbe41
  hotfix
  main
  release
* squash-result
sharafkamaal@192 kamaal123 % git merge --squash feature-from-8ccbe41
Updating 8ccbe41..c669c5f
Fast-forward
Squash commit -- not updating HEAD
sharafkamaal@192 kamaal123 % git log --oneline --graph --all
git status
* c669c5f (feature-from-8ccbe41) JIRA-123 Added login
* 1e825a3 JIRA-123 Added login
* be83814 Added login
* 8ccbe41 (HEAD -> squash-result, origin/feature-from-8ccbe41, main) Reapply "adding a file"
| * d05c297 (tag: v1.0) Add login
|/
* 74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
* cb879dc cherypick
* 2b99f1b doing ammend task
* a9e286a (origin/main, origin/HEAD) Revert "wrong commit" echo "correcting the fix" This reverts commit 6146424f8ec5e1c562767f320f78
04e1cc37ad9f.
* 6146424 wrong commit
```

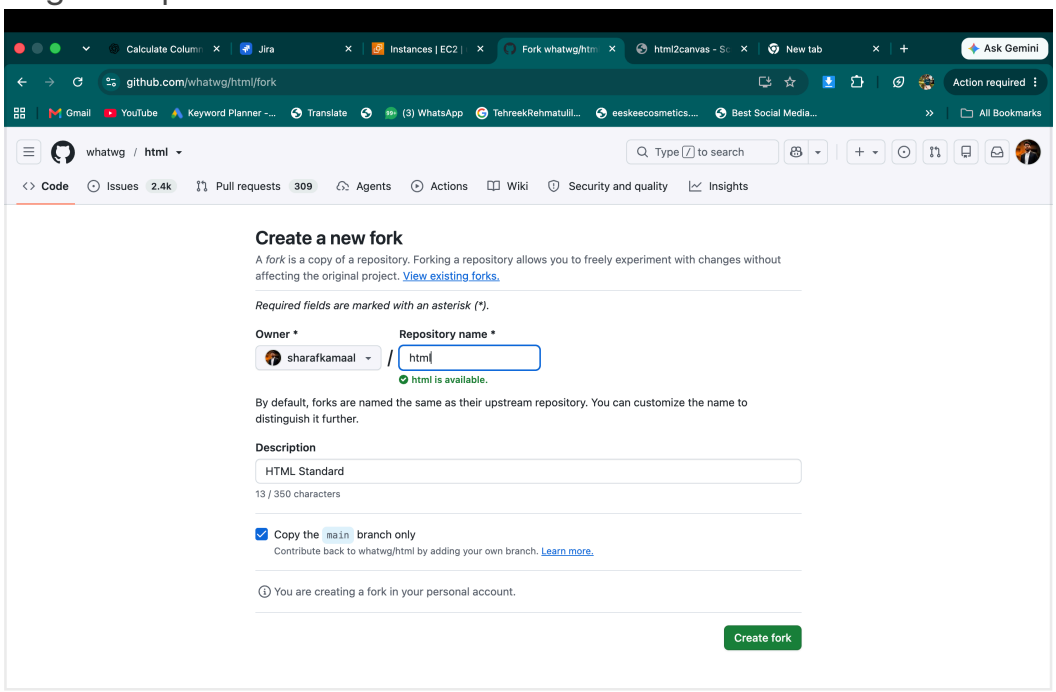



```
* c669c5f (feature-from-8ccbe41) JIRA-123 Added login
* 1e825a3 JIRA-123 Added login
* be83814 Added login
* 8ccbe41 (HEAD -> squash-result, origin/feature-from-8ccbe41, main) Reapply "adding a file"
| * d05c297 (tag: v1.0) Add login
|/
* 74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
* cb879dc cherypick
* 2b99f1b doing ammend task
* a9e286a (origin/main, origin/HEAD) Revert "wrong commit" echo "correcting the fix" This reverts commit 6146424f8ec5e1c562767f320f78
04e1cc37ad9f.
* 6146424 wrong commit
* ca351d8 checking merge conflict1
* 4a3f485 Creating two files
* 8f1f5a0 hiding password file
*   04b63f8 Merge pull request #1 from sharafkamaal/hotfix
|\
| * 241bc05 (origin/hotfix, hotfix) adding a file
|/
| *   bfc9ca6 (release) resolve the conflict issue
| |\
| | * 25b63cc checking merge conflict1
| * | 5feec34 checking merge conflict on release branch
| |/
| * 9d940b4 Creating two files
| | * fe819c5 (origin/release) mergin the files
| | * 5d3579a created a release branch and sample file
| |/
|/|
* | d716abc Creating two files
|/
* 2c23cac Initial commit
On branch squash-result
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store

nothing added to commit but untracked files present (use "git add" to track)
sharafkamaal@192 kamaal123 %
```

```
sharafkamaal@192 kamaal123 % git switch release
Switched to branch 'release'
sharafkamaal@192 kamaal123 % ls
alpha          file.txt       file1.txt      file2.txt      password       README.md
sharafkamaal@192 kamaal123 % git rebase main
warning: skipped previously applied commit 5feec34
warning: skipped previously applied commit 25b63cc
hint: use --reapply-cherry-picks to include skipped commits
hint: Disable this message with "git config set advice.skippedCherryPicks false"
dropping 9d940b49d1441e8ca1acec92fd1bfacd2d3b9292 Creating two files -- patch contents already upstream
Successfully rebased and updated refs/heads/release.
sharafkamaal@192 kamaal123 % git log --oneline --graph --all
* c669c5f (HEAD -> release, main, feature-from-8ccbe41) JIRA-123 Added login
* 1e825a3 JIRA-123 Added login
* be83814 Added login
* 8ccbe41 (origin/feature-from-8ccbe41, squash-result) Reapply "adding a file"
| * d05c297 (tag: v1.0) Add login
|/
* 74b006f Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
* cb879dc cherypick
* 2b99f1b doing ammend task
* a9e286a (origin/main, origin/HEAD) Revert "wrong commit" echo "correcting the fix" This reverts commit 6146424f8ec5e1c562767f320f78
04e1cc37ad9f.
* 6146424 wrong commit
* ca351d8 checking merge conflict1
* 4a3f485 Creating two files
* 8f1f5a0 hiding password file
*   04b63f8 Merge pull request #1 from sharafkamaal/hotfix
|\
| * 241bc05 (origin/hotfix, hotfix) adding a file
|/
| * fe819c5 (origin/release) mergin the files
| * 5d3579a created a release branch and sample file
|/
* d716abc Creating two files
* 2c23cac Initial commit
sharafkamaal@192 kamaal123 % git branch --show-current
release
sharafkamaal@192 kamaal123 % git reflog -10
```

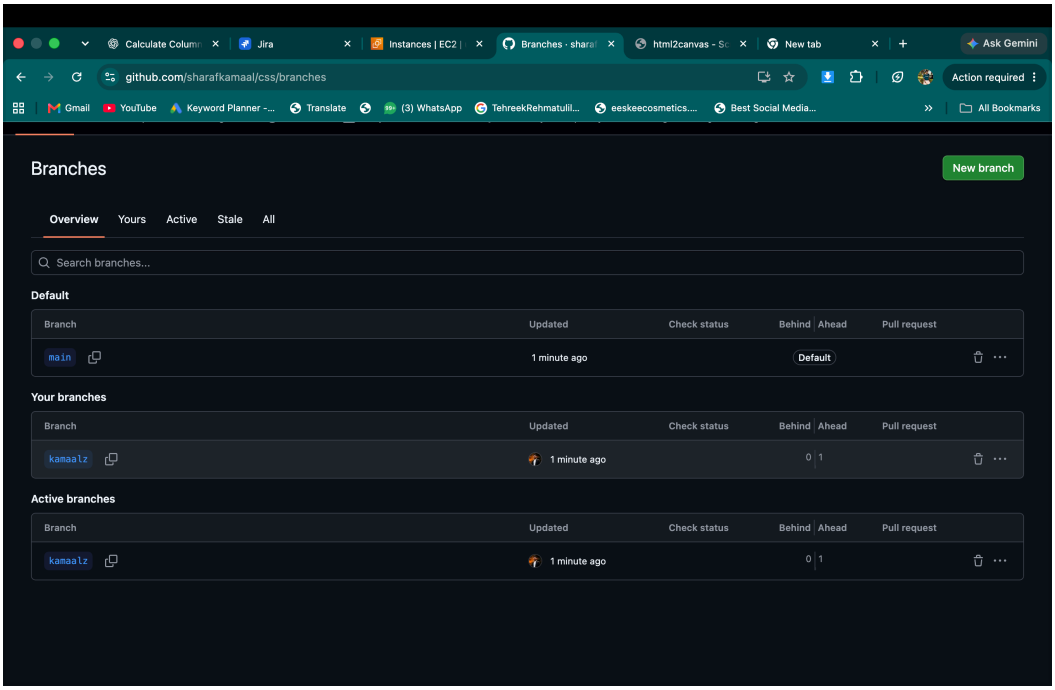
13.Fork & Pull Request Workflow

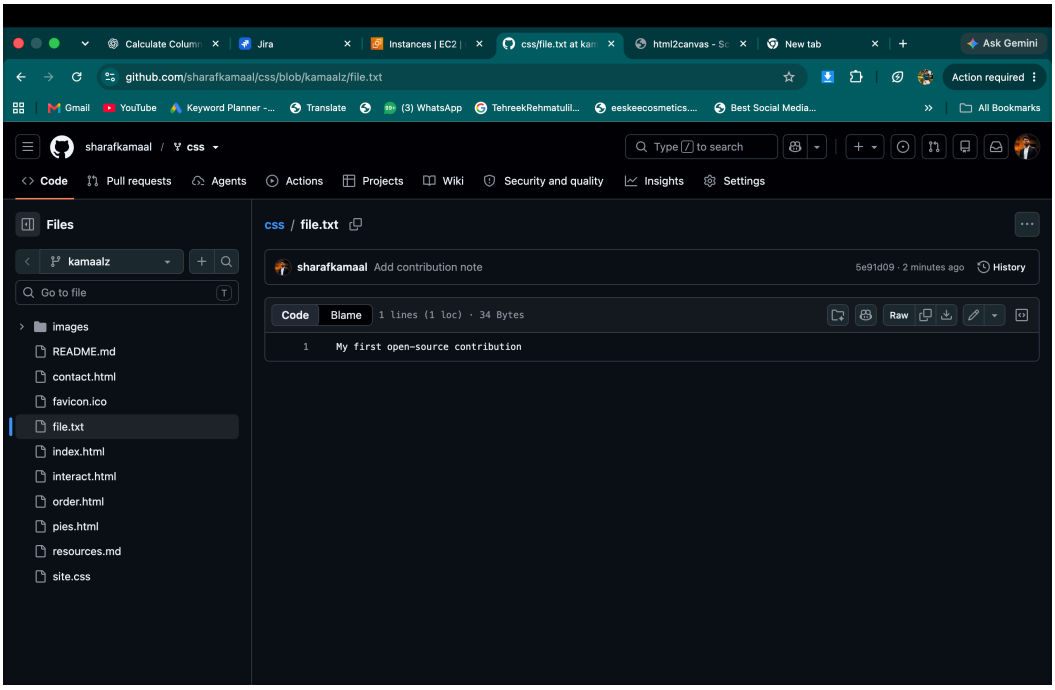
- Fork a repo, make a change, and submit a pull request to the original repo.



```
sharafkamaal@192 devops-techiehorizon % git clone https://github.com/sharafkamaal/html.git
Cloning into 'html'...
remote: Enumerating objects: 49920, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
^Cfetch-pack: unexpected disconnect while reading sideband packet

sharafkamaal@192 devops-techiehorizon % git clone https://github.com/sharafkamaal/html.git
Cloning into 'html'...
remote: Enumerating objects: 49920, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (3/3), done.
^Cfetch-pack: unexpected disconnect while reading sideband packet

sharafkamaal@192 devops-techiehorizon % git clone https://github.com/sharafkamaal/css.git
Cloning into 'css'...
remote: Enumerating objects: 20, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 20 (delta 1), reused 1 (delta 1), pack-reused 17 (from 1)
Receiving objects: 100% (20/20), 541.74 KiB | 1.70 MiB/s, done.
Resolving deltas: 100% (3/3), done.
sharafkamaal@192 devops-techiehorizon % cd css
sharafkamaal@192 css % git switch -c kamaalz
Switched to a new branch 'kamaalz'
sharafkamaal@192 css % echo "My first open-source contribution" > file.txt
sharafkamaal@192 css % git add .
sharafkamaal@192 css % git commit -m "Add contribution note"
[kamaalz 5e91d09] Add contribution note
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
sharafkamaal@192 css % git push -u origin kamaalz
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
```

```
 create mode 100644 file.txt
sharafkamaal@192 css % git push -u origin kamaalz
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 310 bytes | 310.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'kamaalz' on GitHub by visiting:
remote:      https://github.com/sharafkamaal/css/pull/new/kamaalz
remote:
To https://github.com/sharafkamaal/css.git
 * [new branch]      kamaalz -> kamaalz
branch 'kamaalz' set up to track 'origin/kamaalz'.
sharafkamaal@192 css %
```





14.Recover Lost Commit

- Commit something, reset hard, and then recover it using git reflog.

```
sharafkamaal@192 css % ls
contact.html    file.txt        index.html      order.html      README.md       site.css
favicon.ico     images          interact.html   pies.html       resources.md
sharafkamaal@192 css % git reflog
5e91d09 (HEAD -> kamaalz, origin/kamaalz) HEAD@{0}: commit: Add contribution note
1e0e052 (origin/main, origin/HEAD, main) HEAD@{1}: checkout: moving from main to kamaalz
1e0e052 (origin/main, origin/HEAD, main) HEAD@{2}: clone: from https://github.com/sharafkamaal/css.git
sharafkamaal@192 css % git reset --hard HEAD~1
HEAD is now at 1e0e052 update Readme.MD
sharafkamaal@192 css % git reflog
1e0e052 (HEAD -> kamaalz, origin/main, origin/HEAD, main) HEAD@{0}: reset: moving to HEAD~1
5e91d09 (origin/kamaalz) HEAD@{1}: commit: Add contribution note
1e0e052 (HEAD -> kamaalz, origin/main, origin/HEAD, main) HEAD@{2}: checkout: moving from main to kamaalz
1e0e052 (HEAD -> kamaalz, origin/main, origin/HEAD, main) HEAD@{3}: clone: from https://github.com/sharafkamaal/css.git
sharafkamaal@192 css % git branch
* kamaalz
  main
sharafkamaal@192 css % ls
contact.html    images          interact.html   pies.html       resources.md
favicon.ico     index.html      order.html      README.md       site.css
sharafkamaal@192 css % git switch -c recovered-commit 5e91d09
Switched to a new branch 'recovered-commit'
sharafkamaal@192 css % ls
contact.html    file.txt        index.html      order.html      README.md       site.css
favicon.ico     images          interact.html   pies.html       resources.md
sharafkamaal@192 css %
```